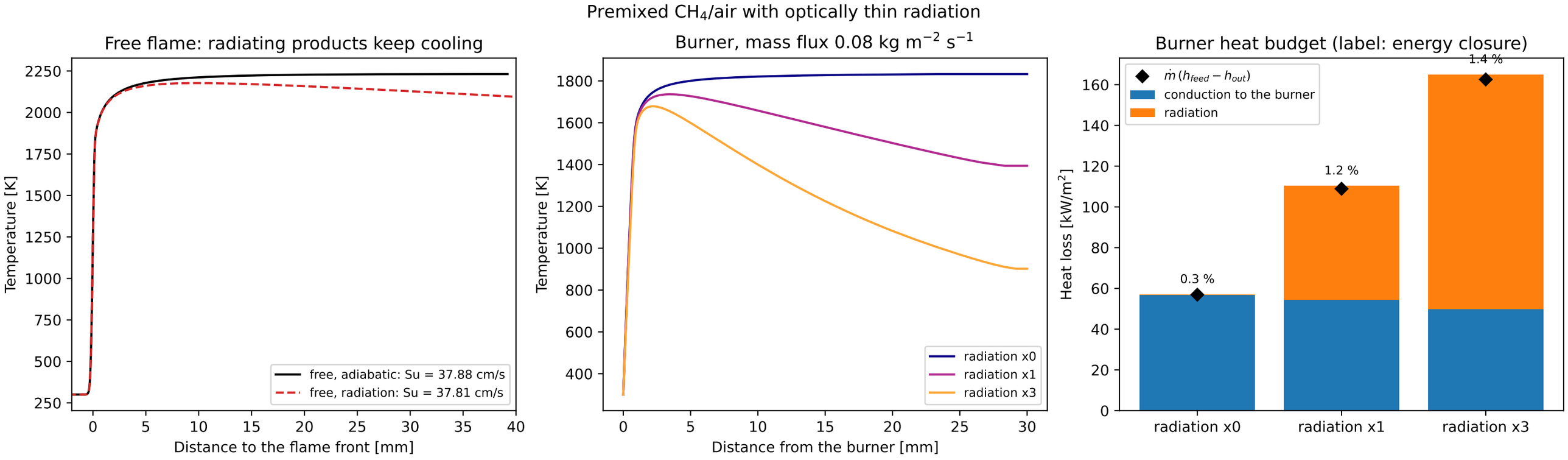
Premixed CH4/air with optically thin radiation
Free flame: radiating products keep cooling
Temperature [K]
Distance to the flame front [mm]
free, adiabatic: Su = 37.88 cm/s
free, radiation: Su = 37.81 cm/s
Burner, mass flux 0.08 kg m−2 s−1
Temperature [K]
Distance from the burner [mm]
radiation x0
radiation x1
radiation x3
Burner heat budget (label: energy closure)
Heat loss [kW/m2]
ṁ (hfeed − hout)
conduction to the burner
radiation
0.3 %
1.2 %
1.4 %
radiation x0
radiation x1
radiation x3